swing your arm back and forth, you are actually swinging all the constituent quarks and electrons back and forth, which brings us to the point. The Higgs ocean in which modern theory claims we are all immersed *interacts*with quarks and electrons: it resists their accelerations much as a vat of molasses resists the motion of a Ping-Pong ball that's been submerged. And this resistance, this drag on particulate constituents, contributes to what you perceive as the mass of your arm and the bowling ball you are swinging, or as the mass of an object you're throwing, or as the mass of your entire body as you accelerate toward the finish line in a 100-meter race. And so we *do* feel the Higgs ocean. The forces we all exert thousands of times a day in order to change the velocity of one object or another—to impart an acceleration—are forces that fight against the drag of the Higgs ocean.[8]

The molasses metaphor captures well some aspects of the Higgs ocean. To accelerate a Ping-Pong ball submerged in molasses, you'd have to push it *much* harder than when playing with it on your basement table—it will resist your attempts to change its velocity more strongly than it does when not in molasses, and so it behaves as if being submerged in molasses has increased its mass. Similarly, as a result of their interactions with the ubiquitous Higgs ocean, elementary particles resist attempts to change their velocities—they acquire mass. However, the molasses metaphor has three misleading features that you should be aware of.

First, you can always reach into the molasses, pull out the Ping-Pong ball, and see how its resistance to acceleration diminishes. This isn't true for particles. We believe that, today, the Higgs ocean fills all of space, so there is no way to remove particles from its influence; all particles have the masses they do regardless of where they are. Second, molasses resists all motion, whereas the Higgs field resists only accelerated motion. Unlike a Ping-Pong ball moving through molasses, a particle moving through outer space at constant speed would not be slowed down by "friction" with the Higgs ocean. Instead, its motion would continue unchanged. Only when we try to speed the particle up or slow it down does the ocean of Higgs field make its presence known by the force we have to exert. Third, when it comes to familiar matter composed of conglomerates of fundamental particles, there is another important source of mass. The quarks constituting protons and neutrons are held